

Antiasthmatic Property of Polyherbal Preparation E-721 B

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E-721B and its aqueous extract were studied for its antiasthmatic effect in various experimental models using rats and guineapigs. E-721B produced significant and dose-dependent inhibition of peritoneal mast-cell degranulation induced by compound 48/80 and egg-albumin in sensitised rats. The inhibition was comparable to that with disodium cromoglycate, ketotifen, and prednisolone. E-721B administered 2 h before the experiment and once daily for seven days significantly increased the preconvulsion time of acetylcholine and histamine aerosol-induced bronchospasm. It also significantly reduced mortality in guineapigs produced by egg albumin-induced anaphylactic shock and also reduced bronchoalveolar fluid eosinophil count in the same animals. E-721B significantly inhibited the acetylcholine and histamine-induced contraction of different isolated smooth muscle preparations from rats and guineapigs. Contrary to these findings E-721B produced dose-dependent contraction of the rat anococcygeus muscle which was blocked by prazosin. Thus our observations establish the antiasthmatic potential of this herbal formulation. Copyright © 2003 John Wiley & Sons, Ltd.

Keywords: antiasthmatics; bronchodilator; *Albizzia lebbeck*; *Solanum xanthocarpum*; *Tylophora indica*; *Glycyrrhiza glabra*; *Achyranthes aspera*.

INTRODUCTION

Bronchial asthma is a chronic respiratory disorder affecting a large proportion of population throughout the world. The currently used drugs for the treatment of this dreadful disease in modern medicine are far from satisfactory as they provide only symptomatic relief, produce several adverse effects and may lose effectiveness on continued use. Hence a number of drugs from indigenous plant sources have been explored for their anti-asthmatic and bronchodilator activities during the last three decades. Some of the promising plants include *Albizzia lebbeck*, *Tylophora indica* and *Glycyrrhiza glabra* etc. (Harnath and Shyamlakumari, 1975; Tripathi and Das, 1977; Geetha *et al.*, 1981). Alkaloidal extract of *Tylophora asthmatica* (Harnath and Shyamlakumari, 1975; Geetha *et al.*, 1981) and water and ethanol extract of *Albizzia lebbeck* have been shown to have antiasthmatic activity (Tripathi and Das, 1977). *Solanum xanthocarpum* has long been reported to be effective in chronic bronchitis, bronchial asthma and non-specific unproductive cough (Bector and Puri, 1971). Recently in asthmatic patients *Solanum xanthocarpum* has been reported to improve pulmonary functions (Govindan *et al.*, 1999).

In the present study, we have investigated the antiasthmatic potential of a multi-drug herbal formulation on various experimental models of bronchial asthma and anaphylaxis. The formulation coded as E-721B consists of *Rhus succidanea* L (Acanthaceae; galls – 20%), *Solanum xanthocarpum* Schood et Wender (Solanaceae; roots – 15%), *Tylophora indica* Bums,

F. (Asclepidiaceae; leaves – 20%), *Albizzia lebbeck* Benth (Leguminosae; bark – 25%), *Glycyrrhiza glabra* L. (Papilionaceae; roots – 5%) and *Achyranthes aspera* L. (Amaranthaceae; leaves – 5%).

MATERIALS AND METHODS

Plant Material Preparation of drug extract

Plant materials were collected from our authorised cultivators in southern and northern parts of India. The plants were identified by botanists of the Himalaya Drug Co. (HDC) Bangalore and compared with a voucher specimen of each plant in the herbarium of their R & D Centre. The dried powders of different plants were mixed in appropriate proportion and this final preparation was supplied to us by HDC, Bangalore. The aqueous extract of the drug was prepared by percolation method using glass percolator. 5 g of the drug was placed in the percolator. 30 ml of distilled water at 50 °C in 3 installments of 10 ml each was added and after 10 min the aliquots were collected in a 100 ml glass beaker. 25 ml of extract was recovered which was subjected to drying under vacuum at 65 °C. The dried extract was then weighed. A yield of 212 ± 5.3 mg (4.25%) was obtained by this method.

Animals and Drugs

Guineapigs of either sex weighing between 350–400 g and albino rats of either sex weighing between 200–250 g were obtained from the animal house of our institute and used throughout the study. E-721B was supplied by Himalaya Drug Company, Bangalore.

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Compound 48/80 (Compound commonly used to induce mast cell degranulation) and Egg albumin were purchased from Sigma (USA) and Glaxo Laboratories, Mumbai, respectively.

(I) Studies on Acetylcholine and Histamine-induced bronchospasm in guineapigs

Guineapigs were selected and randomly divided into six groups each containing six animals. The drugs were administered orally in 0.5% sodium carboxymethyl cellulose (CMC). The single dose treatments were given one and half hour before the study. The following schedule of treatment was administered:

Group I	0.5% CMC (Control)
Group II	Prednisolone (10 mg/kg)
Group III	Ketotifen (1 mg/kg)
Group IV	Aqueous extract of E-721B (250 mg/kg)
Group V	Aqueous extract of E-721B (500 mg/kg)
Group VI	Aqueous extract of E-721B (250 mg/kg) for 7 days.

Later the animals were exposed to an aerosol of 0.25% histamine hydrochloride and time for preconvulsion state was noted for each animal as described by Sheth *et al.* (1972). After about 15 days of wash-out period the same animals were again given the above treatments and preconvulsion time was noted for 0.5% acetylcholine bromide aerosol spray.

(II) Study of E-721B in anaphylactic shock induced mortality in guineapigs

Guineapigs were divided into two groups (control and test) of nine each and were injected with 1 ml of 10% egg albumin subcutaneously and 1 ml of 10% egg albumin intraperitoneally for 3 days. After 4 weeks, the test group animals received E-721B (500 mg/kg, p.o.) as single dose. The control group received only 0.5% C.M.C. After two hours of drug treatment the animals were challenged with 2 ml of 10% egg albumin intraperitoneally and were kept under observation for next 24 h. Any mortality in these animals was noted if it occurred within 24 h.

(III) Isolated tissue experiment

(a) Guinea pig ileum and Guinea pig tracheal chain.

Guineapigs were fasted for 24 h and later were sacrificed and a piece of ileum as well as trachea were isolated. The tissues were quickly transferred to Petri dishes containing Tyrode solution and Krebs's bicarbonate solution respectively. The tracheal chain was prepared and mounted in an organ bath maintained at 37 °C and containing Krebs's bicarbonate solution according to the method described by Sheth *et al.* (1972). The ileum was mounted in an organ bath maintained at 37 °C and containing Tyrode solution. A basal tension of 500 mg was applied to both the tissues throughout the experiment. After stabilization for 45 min the tissues were exposed to graded doses of acetylcholine and/or histamine and contractions were recorded. The responses to these standard drugs were re-elicited after exposing

the tissue to aqueous extract of E-721B for 3 min. After eliciting the response the tissues were washed-out thoroughly before proceeding for next response.

(b) Rat ileum. Albino rats were fasted overnight. The next day the animals were killed and a small piece of ileum was isolated and mounted in an organ bath containing Tyrode solution maintained at 37 °C. A basal tension of 500 mg was applied and the tissue was stabilized for 30 min. The tissue was then exposed to graded doses of acetylcholine and contractions were recorded as described by Sheth *et al.* (1972). The responses to acetylcholine were re-elicited after exposing the tissue to aqueous extract of E-721B for 3 min. After eliciting the response the tissues were washed-out thoroughly before proceeding for next response.

(c) Rat anococcygeus muscle. Albino rats were sacrificed and the anococcygeus muscles were isolated (Gillespie, 1972). They were then suspended in an organ bath containing Krebs's bicarbonate solution maintained at 37 °C. A basal tension of 500 mg was applied and the tissue was stabilized for 30 min. The tissue was then exposed to graded doses of aqueous extract of E-721B. The responses to E-721B were re-elicited after exposing the tissue to prazosin (1×10^{-9} M and 2×10^{-9} M) for 3 min. After eliciting the response the tissues were washed-out thoroughly before proceeding for next response.

(IV) Studies on compound 48/80 and egg albumin induced rat peritoneal mast cell degranulation

A modification of the method described by Kanemoto *et al.* (1993) was employed for the study. Albino rats were sacrificed by stunning and cutting the neck blood vessels. The peritoneal cavity was lavaged with 10 ml of Tyrode solution. The lavaged fluid was collected and centrifuged at 2000 rpm for 5 min. The pellet was separated, washed with Tyrode solution and finally resuspended in 1 ml Tyrode solution. 0.1 ml of this lavage fluid was transferred to 8 test tubes. The lavage fluid was then subjected to following treatment schedule.

Test tube no. 1	Saline
Test tube no. 2	Saline
Test tube no. 3	1 µg/ml of Aq. Extract of E-721B
Test tube no. 4	10 µg/ml of Aq. Extract of E-721B
Test tube no. 5	100 µg/ml of Aq. Extract of E-721B
Test tube no. 6	10 µg/ml of Ketotifen fumarate
Test tube no. 7	10 µg/ml of prednisolone
Test tube no. 8	10 µg/ml of disodium cromoglycate

Each test tubes were incubated for 10 min at 37 °C and then compound 48/80 (0.1 ml, 10 µg/ml) was added to each test tubes except test tube no. 1. After further incubation for 10 min at 37 °C, 0.1 ml of 10% toluidine blue was added and examined under light microscope with 450 × magnification. A minimum of 200 cells was counted for intact and disrupted mast cells and from it % protection from degranulation was calculated.

In another study rats were sensitized by administering egg albumin (1 ml, 10% w/v) intraperitoneally as well as subcutaneously on the first, third, fifth and on the twelfth day of first egg albumin administration. The animals were administered the drugs as per following schedule:

Group I	0.5% CMC Sodium
Group II	Prednisolone (1 mg/kg)
Group III	E-721B (50 mg/kg)
Group IV	E-721B (100 mg/kg)
Group V	E-721B (200 mg/kg)

Two hours later the mast cell were isolated and the above study was conducted in the same manner as above with the one single change of replacement of compound 48/80 by 0.1 ml 1% egg albumin.

(V) Bronchoalveolar lavage study

The guineapigs, with the exception of the plain control (unsensitized animal) group, were sensitized 15 days before test with egg albumin (1 ml, 10% w/v) intraperitoneally. On the day of experiment the drug E-721B (500 mg/kg) was administered orally as suspension in 0.5% CMC to test animals. Control (sensitized guineapigs treated with vehicle) as well as the plain control animals received only 0.5% CMC. Two hours after drug administration animals were challenged with 0.5 ml of 2% egg albumin intravenously through saphenous vein. After 3 h of the challenge of the egg albumin or just prior to death of the animals, whichever was earlier, the tracheobronchial tree was lavaged with 10 ml of saline as described by Thomas *et al.* (1995) and the fluid so collected was centrifuged at 2000 rpm for 5 min and the pellet was resuspended in 0.5 ml saline. 0.2 ml of Geimsa stain in buffered saline (6.8 pH) was added to it. After 5 min the number of each type of leucocyte in 0.05 ml fluid was determined under the microscope using 450 × magnification.

Statistical Method

All the results were tested for significance using paired Student's *t*-test at the probability level of 95%.

RESULTS

(I) Effect of E-721B on Acetylcholine and Histamine-induced bronchospasm in guinea pigs

Significant increase in preconvulsion time was observed due to pretreatment with E-721B (250 mg/kg and 500 mg/kg P.O.) when the guineapigs were exposed to either acetylcholine (0.5%) or histamine (0.25%) aerosol. The increase in preconvulsion time was greater with one week of once daily treatment with E-721B (250 mg/kg) than single dose treatment with either E-721B (250 mg/kg) or E-721B (500 mg/kg). The spasmolytic effect of E-721B was comparable to both prednisolone (10 mg/kg) and ketotifen (1 mg/kg). (Table 1).

(II) Effect of E-721B on anaphylactic shock induced mortality in guineapigs

The animals not subjected to drug pretreatment showed 100% mortality within 10 h after egg albumin challenge. However, E-721B (500 mg/kg, p.o.) administered 2 h

Table 1. Effect of E-721B on Acetylcholine and Histamine induced bronchospasm in guinea pigs

Sr.no.	Treatment	% Increase in preconvulsion time	
		Acetylcholine	Histamine
1	E-721B (250 mg/kg)	17.9 ± 3.68	28.8 ± 2.77
2	E-721B (500 mg/kg)	30.8 ± 6.74	38.6 ± 2.69
3	E-721B (250 mg/kg) (Once daily for 7 days)	35.1 ± 4.76	47.1 ± 3.20
4	Prednisolone (10 mg/kg)	36.2 ± 3.89	50.6 ± 2.88
5	Ketotifen (1 mg/kg)	16.4 ± 4.36	33.9 ± 3.16

Values are mean ± SEM, *n* = 6 in each group.

* *p* < 0.05 (Student's paired *t*-test).

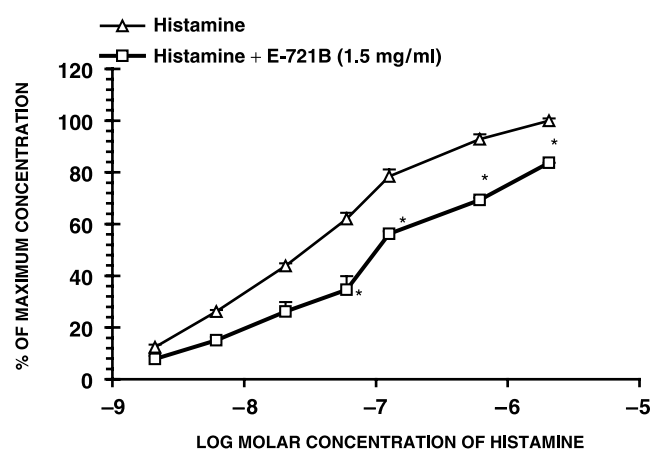


Figure 1. Effect of E-721B on histamine induced contractions of guineapig ileum. Each point and the bar represent mean ± SEM of six observations. * *p* < 0.05 paired students *t* test.

prior to egg albumin challenge significantly reduced the mortality (63%) even when observed up to 24 h.

(III) Effect of E-721B on isolated tissues

(a) Effect of E-721B on isolated guineapig ileum. Histamine (2.13×10^{-9} M to 2.06×10^{-6} M) as well as acetylcholine (1.62×10^{-7} M to 1.62×10^{-5} M) produced dose-dependent contractions of guineapig ileum. Pretreatment with aqueous extract of E-721B (1.5 mg/ml) significantly inhibited the contractions to histamine as well as acetylcholine (Figs 1 and 2).

(b) Effect of E-721B on rat ileum. Acetylcholine (2.54×10^{-7} M to 1.3×10^{-5} M) produced a dose-dependent contractions of rat ileum. Pretreatment with aqueous extract of E-721B (1.5 mg/ml) significantly inhibited the contractions to acetylcholine (Fig. 3).

(c) Effect of E-721B on guineapig tracheal chain. Histamine (2.10×10^{-8} M to 2.10×10^{-6} M) produced dose-dependent contractions of guineapig tracheal chain. Pretreatment with aqueous extract of E-721B

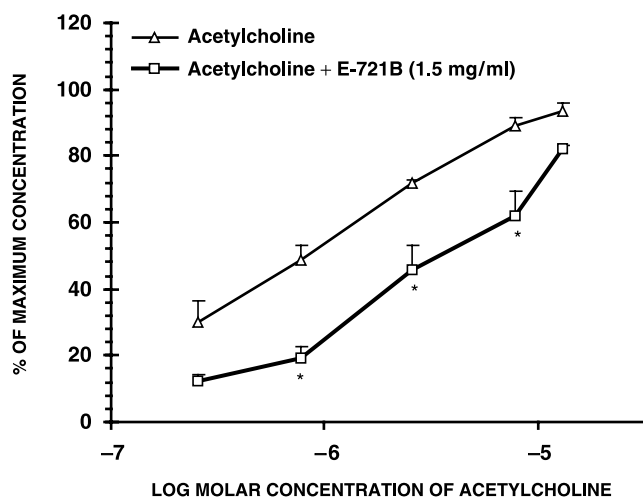


Figure 2. Effect of E-721B on acetylcholine induced contractions of guineapig ileum. Each point and the bar represent mean $\pm$ SEM of six observations. * $p < 0.05$ paired students t test.

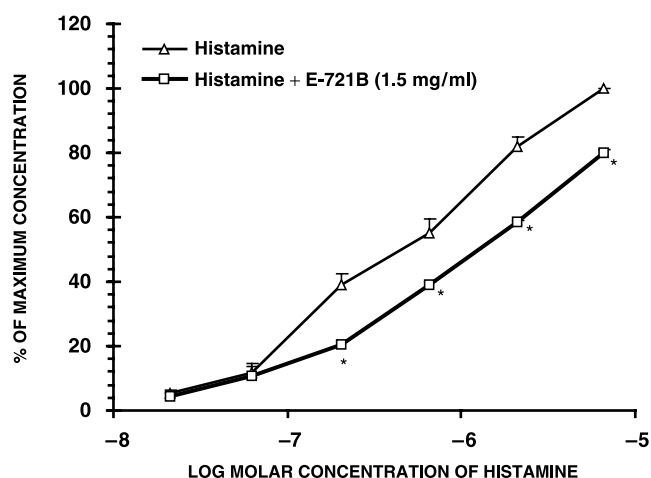


Figure 4. Effect of E-721B on histamine induced contractions of guineapig tracheal chain. Each point and the bar represent mean $\pm$ SEM of six observations. * $p < 0.05$ paired students t test.

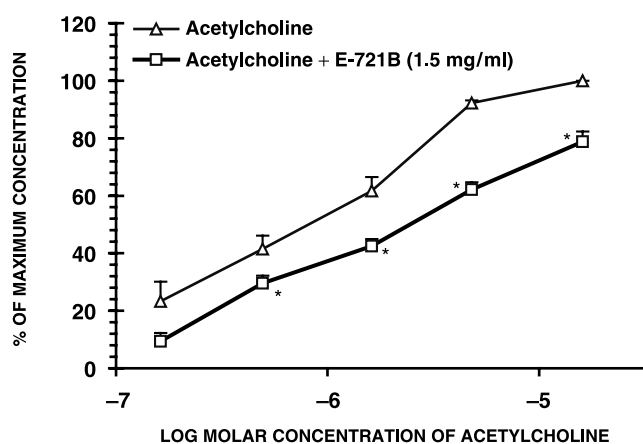


Figure 3. Effect of E-721B on acetylcholine induced contractions of rat ileum. Each point and the bar represent mean $\pm$ SEM of six observations. * $p < 0.05$ paired students t test.

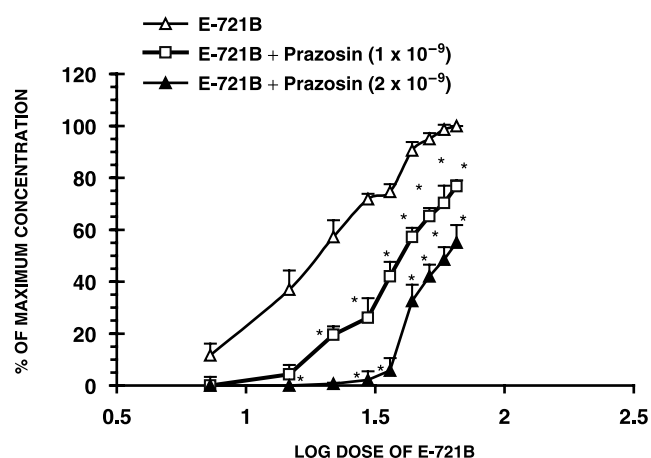


Figure 5. Effect of E-721B and its interaction with prazosin on rat anococcygeus muscle. Each point and the bar represent mean $\pm$ SEM of six observations. * $p < 0.05$ paired students t test.

(1.5 mg/ml) significantly inhibited the contractions to histamine (Fig. 4).

(d) Effect on rat anococcygeus muscle. Dose dependent contractions to aqueous extract of E-721B (7.27 mg/ml to 65.21 mg/ml) were observed and pretreatment with prazosin significantly inhibited the contractions to aqueous extract of E-721B (Fig. 5).

(IV) Effect of E-721B on rat peritoneal mast cell degranulation

Compound 48/80 produced significant disruption of mast cells which was significantly inhibited by pretreatment with the aqueous extract of E-721B in concentrations of 10 μ g/ml and 100 μ g/ml. The protection was comparable to the standard drugs prednisolone (10 μ g/ml), ketotifen (10 μ g/ml) and disodium cromoglycate (10 μ g/ml) (Table 2).

Egg albumin induced mast cell degranulation was significantly inhibited in a dose-dependent manner by pretreatment of the animals with aqueous extract of

Table 2. Effect of E-721B on compound 48/80 induced mast cell degranulation

Sr.no.	Treatment	% Protection from Degranulation
1	Control	76.4 $\pm$ 4.89
2	C48/80 (10 μ g/ml)	20.2 $\pm$ 6.63
3	C48/80 (10 μ g/ml) Prednisolone (10 μ g/ml)	65.2 $\pm$ 8.76
4	C48/80 (10 μ g/ml) Ketotifen (1 μ g/ml)	50.6 $\pm$ 6.64
5	C48/80 (10 μ g/ml) Cromoglycate (10 μ g/ml)	60.1 $\pm$ 6.38
6	C48/80 (10 μ g/ml) E-721B (1 μ g/ml)	33.7 $\pm$ 4.33
7	C48/80 (10 μ g/ml) E-721B (1 μ g/ml)	43.8 $\pm$ 3.97
8	C48/80 (10 μ g/ml) E-721B (1 μ g/ml)	59.6 $\pm$ 3.34

Values are mean $\pm$ SEM, $n = 6$ in each group.

* $p < 0.05$ (Student's paired t -test).

Table 3. Effect of E-721B on egg albumin induced rat peritoneal mast cell degranulation

Sr.no.	Treatment	% Protection from Degranulation
1	Control	20.8 ± 2.21
2	Prednisolone (10 mg/kg)	73.3 ± 5.68
3	E-721B (50 mg/kg)	22.3 ± 2.87
4	E-721B (100 mg/kg)	44.63 ± 4.23
5	E-721B (200 mg/kg)	58.9 ± 3.77

Values are mean ± SEM, $n = 6$ in each group.

* $p < 0.05$ (Student's paired t -test).

E-721B. The effect, though less, was comparable to prednisolone-induced protection (Table 3).

(V) Effect of E-721B on differential leucocyte counts of bronchial fluid in egg albumin sensitised guineapigs

As compared with unsensitised animals (plain control group), significant increases in eosinophil and basophil count were observed in untreated sensitised animals (control group) challenged with egg albumin. However, the effect on the other leucocytes was not significant. No significant difference was observed in the count of all the types of WBC detected in the bronchial fluid of sensitised animals pretreated with E-721B and unsensitised-untreated animals (Table 4).

DISCUSSION

The results of this study reveal significant protection against bronchoconstriction, anti-anaphylactic and anti-spasmodic activity of E-721B and its aqueous extract on various in vivo and in vitro experimental models used by us.

E-721B significantly delayed the onset of convulsions in guineapigs due to acute bronchospasm induced by acetylcholine and histamine aerosols. It was more effective after chronic administration for 7 days in these models. These observations substantiate its protective effect against bronchoconstriction. Similar results have been reported with *Albizzia lebeck* and *Glycyrrhiza glabra* (Tripathi and Das, 1977) which are constituents of E-721B and may be responsible for

preventing the bronchospasm induced by histamine and acetylcholine. E-721B, in a single dose of 500 mg/kg, also produced significant reduction in the mortality of guineapigs subjected to anaphylactic shock induced by egg albumin. Significant protection of guineapigs against horse serum induced anaphylaxis has been recently reported for this formulation following its chronic administration for 14 days (Mitra *et al.*, 1999). Individually *Albizzia lebeck* and *Tylophora indica* have also been shown to possess anti-anaphylactic activity in rats and guineapigs (Tripathi and Das, 1977; Nayampalli and Sheth, 1979).

A significant protection of rat peritoneal mast cells from disruption caused by antigen and compound 48/80 in our study and after active and passive anaphylaxis (Tripathi *et al.*, 1979) in rats points towards the role of mast cell stabilization in the possible mechanism of action of E-721B. *A. lebeck* and *T. indica* have also been shown to produce mast cell stabilization in earlier studies (Tripathi *et al.*, 1979; Geetha *et al.*, 1981). *A. lebeck* inhibits the sensitization process and synthesis of reaginic type humoral antibodies in sensitized peritoneal mast cells of albino rats when challenged with antigen (Tripathi *et al.*, 1979) and both *A. lebeck* and *T. indica* also possess a disodium cromoglycate like mast cell stabilization property probably by raising the cyclic AMP levels due to inhibition of the enzyme phosphodiesterase (Geetha *et al.*, 1981).

Recently the role of eosinophils in allergic conditions has been highlighted (Dunn *et al.*, 1988; Ishida *et al.*, 1989; Sanjar *et al.*, 1990a). Exposure of actively sensitized guineapigs to allergens result in selective pulmonary eosinophilia, a response that has been associated with airway hyperreactivity (Santing *et al.*, 1994; Sanjar *et al.*, 1990a). An association between the presence of an eosinophilic infiltrate and airways hyperreactivity in the asthmatic lung has led to the concept that hyperreactivity was secondary to, and therefore, a consequence of pulmonary inflammation (Barnes, 1986). Exposure to PAF (Platelet Activating Factor) aerosols has also been shown to produce selective accumulation of eosinophils in bronchoalveolar fluid in guineapigs (Sanjar *et al.*, 1990b). Our observations showing a significant reduction caused by E-721B in eosinophil counts in bronchoalveolar fluid in guineapigs subjected to anaphylactic shock points towards a possibility of antagonism to endogenous PAF and/or other mediators of anaphylaxis and inflammation which in turn may be responsible for protection against the antigenic response of sensitized animals.

Table 4. Effect of E-721B on change in leucocyte count due to egg albumin induced anaphylaxis in guineapigs

Sr.no.	Type of W.B.C.	% increase from initial count	
		Control	E-721B (500 mg/Kg, i.p.)
1	Monocytes	0.80 ± 0.10	-1.0 ± 0.12
2	Eosinophils	103.3 ± 10.76	-3.4 ± 0.17*
3	Basophils	93.5 ± 7.89	10.36 ± 1.47*
4	Neutrophils	-3.63 ± 1.77	2.67 ± 3.03
5	Lymphocytes	7.83 ± 4.76	6.36 ± 5.33

Values are mean ± SEM, $n = 6$ in each group.

* $p < 0.05$ (Student's paired t -test).

Our studies on the isolated tissues like guineapig tracheal chain and ileum, and rat ileum substantiate the H₁ antihistaminic and antimuscarinic activity of E-721B whereas observations on the rat anococcygeus muscle points towards a direct sympathomimetic action on alpha-1 receptors as the response is blocked by prazosin.

In conclusion E-721B possesses significant protection against bronchoconstriction and anti-anaphylactic activity. Stabilization of mast cells and inhibition of pulmonary eosinophilia appear to be involved in the unique mode of action. Other factors, which may add to its activity, are its H₁ antihistaminic, antimuscarinic and direct sympathomimetic activities observed in our study.

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